



DeepWalk: Online Learning of Social Representations

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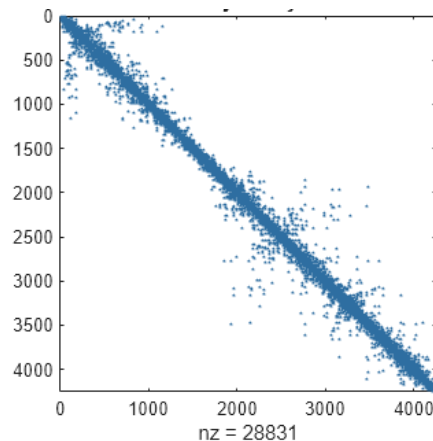
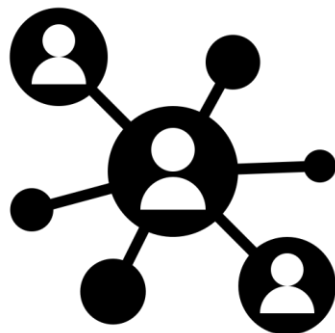
Dayune Choi

- Introduction
- DeepWalk
- Methodology
- Experiment
- Conclusion

The Sparsity of a network

□ Network

- A collection of nodes and the links that connect them
- In real-world networks, nodes are not uniformly connected, resulting in an inherently sparse structure
- Sparsity poses a significant obstacle to model generalization



Limitation of Traditional ML on Networks

□ Traditional Classification

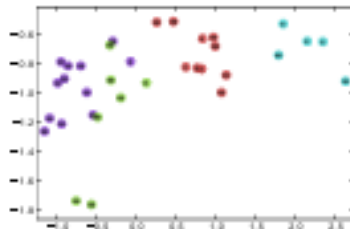
- To learn $H : X \rightarrow Y$
- Assumes examples are independent and identically distributed

□ The Reality of Networks

- Networks possess significant information embedded in their structure
- Links violate the independence assumption, as a node's identity is defined by its neighbors



(a) Input: Karate Graph



(b) Output: Representation

☐ Relational Classification

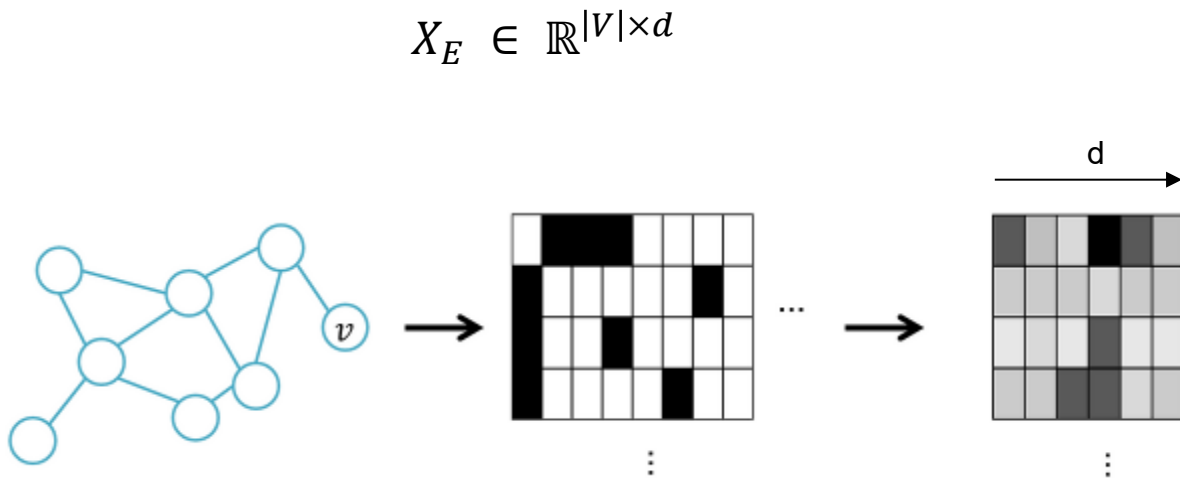
- To infer labels by exploiting dependencies among linked instances
- Traditional Approaches : Use inference techniques

☐ Limitations

- Cascading Error
- Computational cost

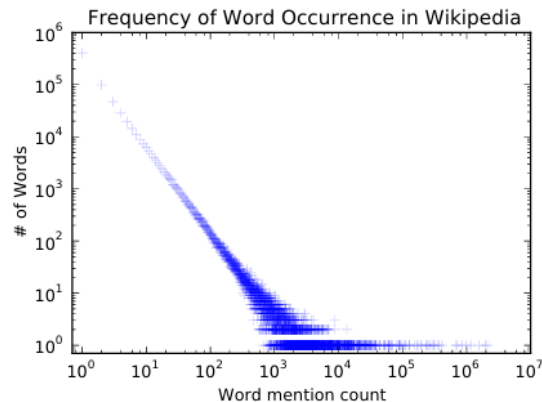
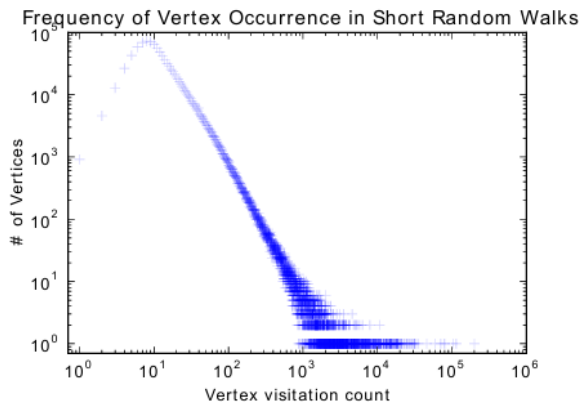
□ Goal

- Transform the network into a dense, continuous vector space



□ Network Distribution : Power Law

- A tiny fraction of nodes monopolize the vast majority of connections, while most nodes have very few
- Just as words form contexts in sentences, nodes form contexts within network structures



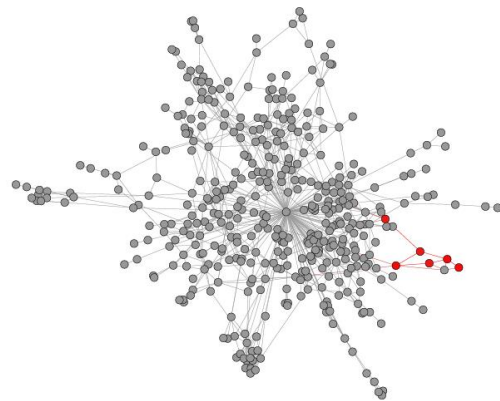
Random Walk as Sentence

□ Random Walk

- A stochastic process that traverses a graph by moving to neighboring nodes
- It enables the capture of both local and higher-order structural context

□ Why Random Walk?

- Random walks naturally reflect imbalance by sampling nodes according to the network topology
- Graph structure $\rightarrow$ Node sequence

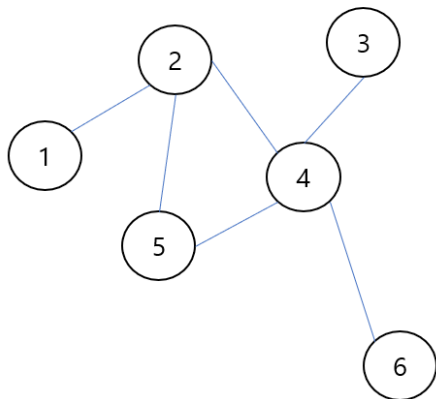


(a) Random walk generation.

Random Walk as Sentence

□ Hyperparameter

- Walk per vertex γ
- Walk length t
- Window size w



❖

Random walk ($t = 6$) starting from node 2

$2 \rightarrow 1 \rightarrow 2 \rightarrow 5 \rightarrow 4 \rightarrow 3$

❖

$w = 2$

Center node : 5

Context node : {1,2,4,3}

❖

Training pairs :

(5,1), (5,2), (5,3), (5,4)

□ Skip-gram in DeepWalk

- To maximize the log-probability of predicting neighboring words given a target word
- Maximize the co-occurrence probability of context nodes given a center node

$$Pr(\{v_{i-w}, \dots, v_{i+w}\} \setminus v_i \mid \Phi(v_i))$$

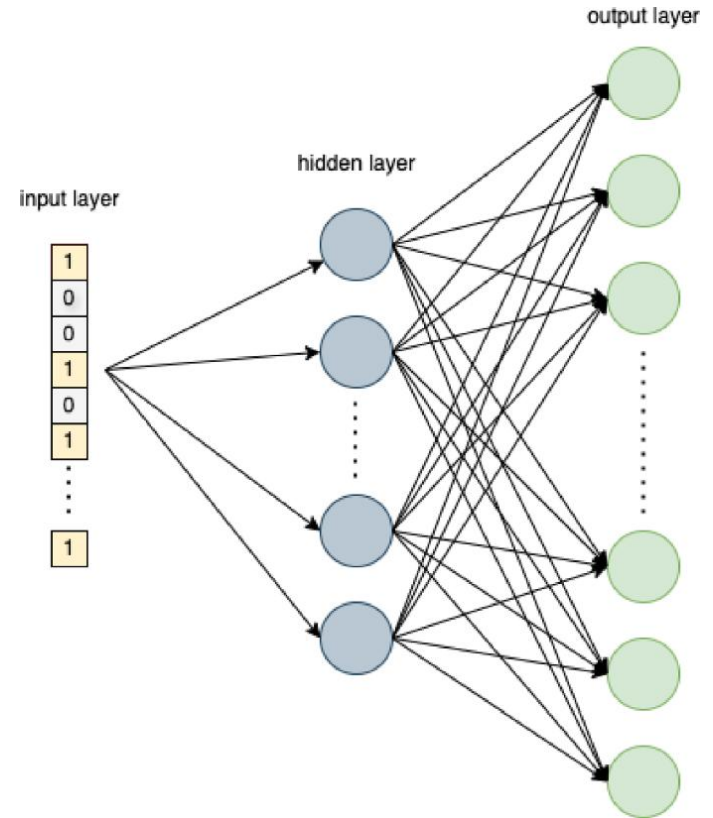
- Assume context nodes are conditionally independent given the center node's representation

$$\prod_{\substack{j=i-w \\ j \neq i}}^{i+w} Pr(v_j \mid \Phi(v_i))$$

Skip-gram

□ Skip-gram architecture

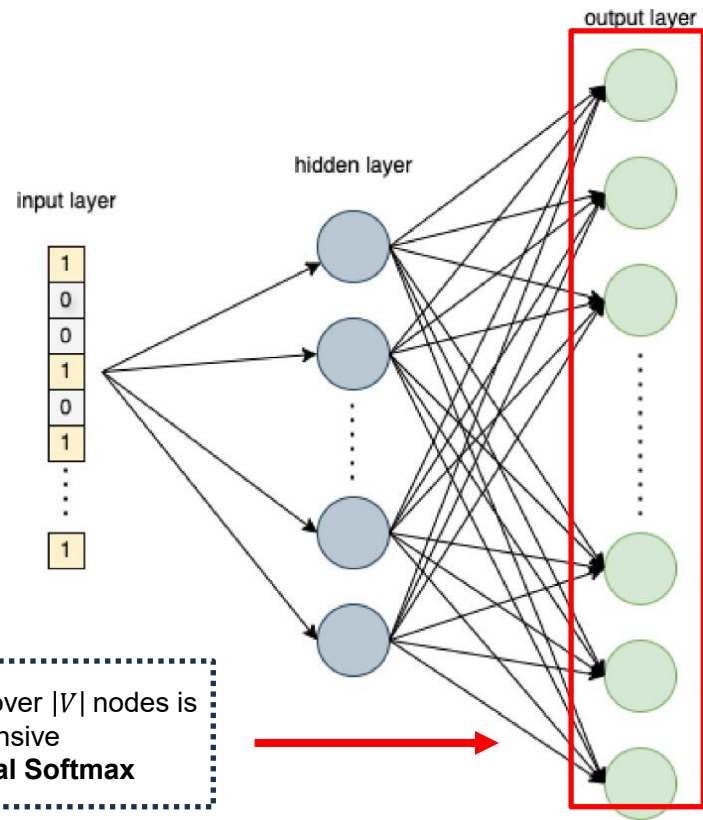
- Input layer : Receive a vector of the center node
- Hidden layer : Map the input to a latent representation
- Output layer : Compute the dot product between the center Node embedding and all vertex



Skip-gram

□ Skip-gram architecture

- Input layer : Receive a vector of the center node
- Hidden layer : Map the input to a latent representation
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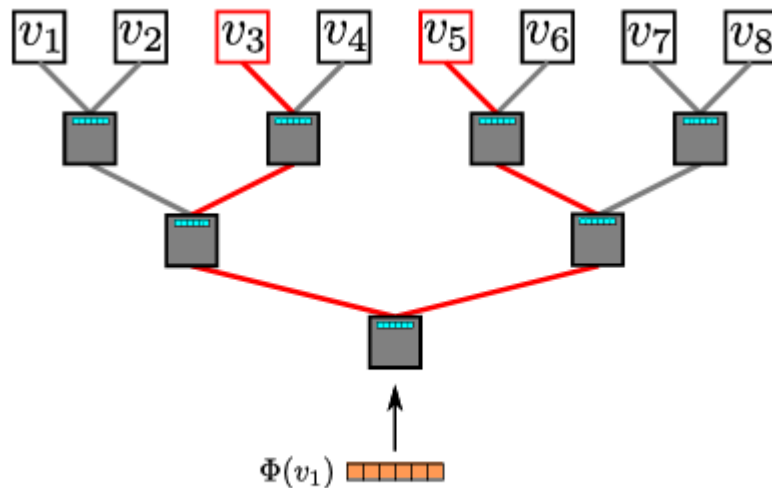


Calculating Softmax over $|V|$ nodes is computationally expensive
: requires **Hierarchical Softmax**

□ Hierarchical Softmax

- Binary Tree Structure: All vertices are assigned to the leaves of a binary tree
- The prediction problem is transformed into maximizing the probability of a specific path from the root to the target leaf
- Each internal node possesses a parameter vector
- Frequent vertices in random walks are assigned shorter paths

$$\Pr(b_l \mid \Phi(v_j)) = 1/(1 + e^{-\Phi(v_j) \cdot \Psi(b_l)})$$



(c) Hierarchical Softmax.

☐ Experimental Design

- Datasets : Blogcatalog, Flickr, YouTube

- Baseline Methods

- ☐ Matrix-based : Spectral Clustering, Modularity
- ☐ Clustering-based : EdgeCluster
- ☐ Neighborhood-based : wvRN
- ☐ Basic : Major

- Experiment

- ☐ Multi-label Classification : Evaluate representation quality under limited labeled data
Test generalization with varying training ratio
- ☐ Parameter Sensitivity : To evaluate how changes to the parameterization of DeepWalk

	% Labeled Nodes	10%	20%	30%	40%	50%	60%	70%	80%	90%
	DEEPWALK	36.00	38.20	39.60	40.30	41.00	41.30	41.50	41.50	42.00
Micro-F1(%)	SpectralClustering	31.06	34.95	37.27	38.93	39.97	40.99	41.66	42.42	42.62
	EdgeCluster	27.94	30.76	31.85	32.99	34.12	35.00	34.63	35.99	36.29
	Modularity	27.35	30.74	31.77	32.97	34.09	36.13	36.08	37.23	38.18
	wvRN	19.51	24.34	25.62	28.82	30.37	31.81	32.19	33.33	34.28
	Majority	16.51	16.66	16.61	16.70	16.91	16.99	16.92	16.49	17.26
	DEEPWALK	21.30	23.80	25.30	26.30	27.30	27.60	27.90	28.20	28.90
Macro-F1(%)	SpectralClustering	19.14	23.57	25.97	27.46	28.31	29.46	30.13	31.38	31.78
	EdgeCluster	16.16	19.16	20.48	22.00	23.00	23.64	23.82	24.61	24.92
	Modularity	17.36	20.00	20.80	21.85	22.65	23.41	23.89	24.20	24.97
	wvRN	6.25	10.13	11.64	14.24	15.86	17.18	17.98	18.86	19.57
	Majority	2.52	2.55	2.52	2.58	2.58	2.63	2.61	2.48	2.62

Table 2: Multi-label classification results in BLOGCATALOG

	% Labeled Nodes	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
Micro-F1(%)	DEEPWALK	32.4	34.6	35.9	36.7	37.2	37.7	38.1	38.3	38.5	38.7
	SpectralClustering	27.43	30.11	31.63	32.69	33.31	33.95	34.46	34.81	35.14	35.41
	EdgeCluster	25.75	28.53	29.14	30.31	30.85	31.53	31.75	31.76	32.19	32.84
	Modularity	22.75	25.29	27.3	27.6	28.05	29.33	29.43	28.89	29.17	29.2
	wvRN	17.7	14.43	15.72	20.97	19.83	19.42	19.22	21.25	22.51	22.73
	Majority	16.34	16.31	16.34	16.46	16.65	16.44	16.38	16.62	16.67	16.71
Macro-F1(%)	DEEPWALK	14.0	17.3	19.6	21.1	22.1	22.9	23.6	24.1	24.6	25.0
	SpectralClustering	13.84	17.49	19.44	20.75	21.60	22.36	23.01	23.36	23.82	24.05
	EdgeCluster	10.52	14.10	15.91	16.72	18.01	18.54	19.54	20.18	20.78	20.85
	Modularity	10.21	13.37	15.24	15.11	16.14	16.64	17.02	17.1	17.14	17.12
	wvRN	1.53	2.46	2.91	3.47	4.95	5.56	5.82	6.59	8.00	7.26
	Majority	0.45	0.44	0.45	0.46	0.47	0.44	0.45	0.47	0.47	0.47

Table 3: Multi-label classification results in FLICKR

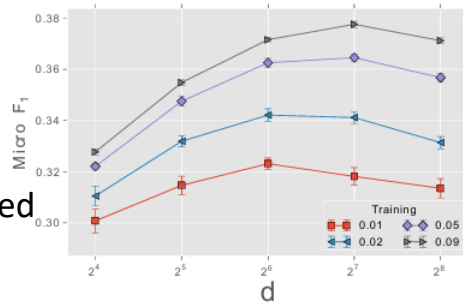
	% Labeled Nodes	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
Micro-F1(%)	DEEPWALK	37.95	39.28	40.08	40.78	41.32	41.72	42.12	42.48	42.78	43.05
	SpectralClustering	—	—	—	—	—	—	—	—	—	—
	EdgeCluster	23.90	31.68	35.53	36.76	37.81	38.63	38.94	39.46	39.92	40.07
	Modularity	—	—	—	—	—	—	—	—	—	—
	wvRN	26.79	29.18	33.1	32.88	35.76	37.38	38.21	37.75	38.68	39.42
	Majority	24.90	24.84	25.25	25.23	25.22	25.33	25.31	25.34	25.38	25.38
Macro-F1(%)	DEEPWALK	29.22	31.83	33.06	33.90	34.35	34.66	34.96	35.22	35.42	35.67
	SpectralClustering	—	—	—	—	—	—	—	—	—	—
	EdgeCluster	19.48	25.01	28.15	29.17	29.82	30.65	30.75	31.23	31.45	31.54
	Modularity	—	—	—	—	—	—	—	—	—	—
	wvRN	13.15	15.78	19.66	20.9	23.31	25.43	27.08	26.48	28.33	28.89
	Majority	6.12	5.86	6.21	6.1	6.07	6.19	6.17	6.16	6.18	6.19

Table 4: Multi-label classification results in YOUTUBE

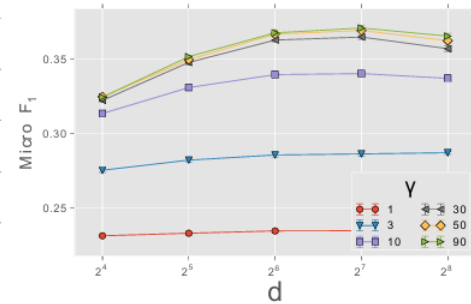
Experiment

□ Effect of Dimensionality

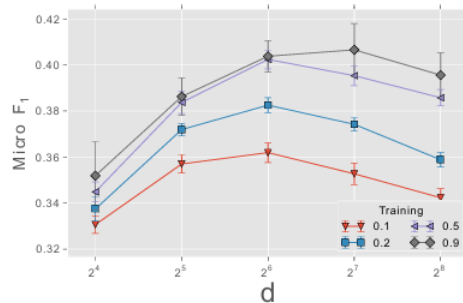
The optimal embedding dimensionality is not fixed
performance consistently degrades beyond
a certain dimension



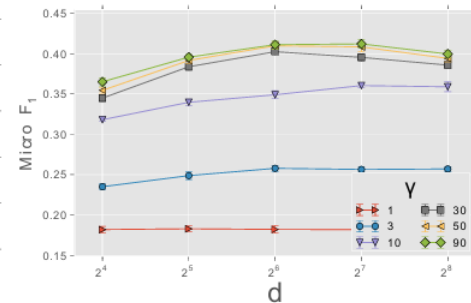
(a1) FLICKR, $\gamma = 30$



(a2) FLICKR, $T_R = 0.05$



(a3) BLOGCATALOG, $\gamma = 30$



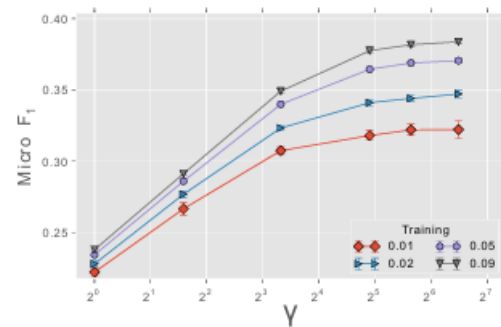
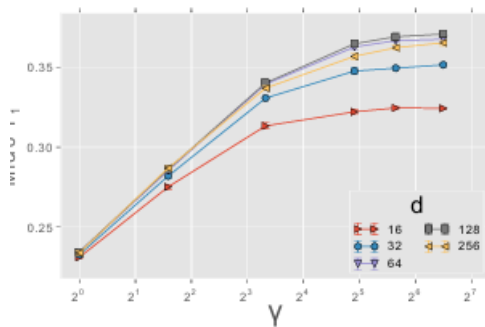
(a4) BLOGCATALOG, $T_R = 0.5$

(a) Stability over dimensions, d

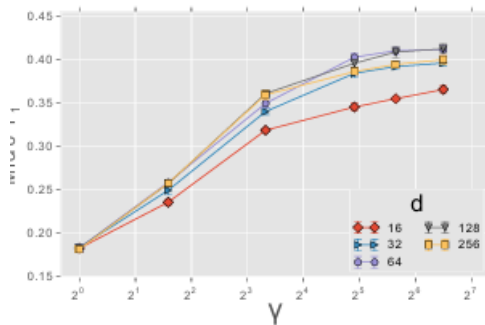
Experiment

□ Effect of Sampling frequency

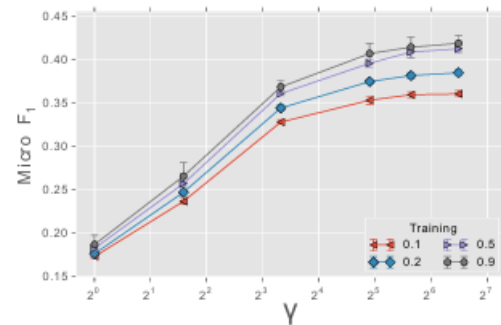
Meaningful embeddings can be learned with only a small number of random walks



(b1) FLICKR, $T_R = 0.05$



(b2) FLICKR, $d = 128$



(b3) BLOGCATALOG, $T_R = 0.5$

(b4) BLOGCATALOG, $d = 128$

Conclusion



- Transform sparse adjacency matrices into Low-dimensional Dense Embeddings to capture latent structural identities
- Defined Nodes as Words and Random Walks as Sentences, enabling the use of proven NLP models for network learning
- Reduce complexity bypassing expensive matrix computations
- Maintaining superior performance even in label-sparse environments